

Daniel Zou | Curriculum Vitae

✉ zoudj@umich.edu

Education

University of Michigan - Ann Arbor

Ph.D. in Statistics

Outstanding First Year Student

Ann Arbor, MI

expected May 2028

Reed College

B.A. in Mathematics - Computer Science, GPA 3.85/4.00

Commendation for Excellence in Scholarship 2019-2020, 2021-2022, 2022-2023, Phi Beta Kappa

Coursework includes: Probability, Mathematical Statistics, Real Analysis, Functional Analysis, Algorithms and Data Structures, Computability and Complexity, Artificial Intelligence, Statistical Learning, Causal Inference

Portland, OR

May 2023

Research Experience

Reed College

Senior Thesis

Portland, OR

September 2022 – May 2023

- Advisors: *Dr. Leonard Wainstein* (Reed College) and *Ira Globus-Harris* (University of Pennsylvania)
- Adapted the Theil-Sen method for weighted linear regression
- Developed an algorithm for differentially private weighted linear regression

Reed College

Differentially Private Statistics REU

Portland, OR

June 2022 – ongoing

- Advisors: *Dr. Adam Groce* (Reed College) and *Dr. Andrew Bray* (UC Berkeley)
- Developed algorithms for faster computation for rank-based statistics
- Reduced sample complexity required for the k-group equality of distributions test by 42%
- Derived higher power private hypothesis tests and change point analysis for 3 other settings

University of Connecticut

Markov Chains REU

Storrs, CT

June – August 2021

- Advisor: *Dr. Iddo Ben-Ari* (University of Connecticut)
- Proved an open conjecture on the expected coalescing time of a coalescing walk on an n-cycle
- Developed algorithms in Python to simulate coalescing walks

Relevant Experience

University of Michigan Statistics Department

Graduate Student Instructor

Ann Arbor, MI

August 2023 – Ongoing

- GSI for STATS 485 Statistics Capstone Seminar
- GSI for STATS 270 Honors Introduction to Statistics
- GSI for STATS 250 Introduction to Statistics

Reed College Mathematics Department

Course Assistant

Portland, OR

August 2021 – May 2023

- Assisted MATH 391 Probability
- Assisted MATH 201 Linear Algebra
- Assisted MATH 141 Introduction to Statistics
- Hosted office hours twice a week to answer questions regarding the homework and course material
- Graded homework and exams

Reed College Computer Science Department

Course Assistant

- Assisted CS 387 Computability and Complexity
- Assisted CS 382 Algorithms and Data Structures
- Hosted office hours twice a week and tutored students individually

Portland, OR

August 2022 – May 2023

Reed College Disability and Accessibility Resources

Specialized Notetaker

- Worked with DAR to provide accommodations via detailed lecture notes for MATH 321 Real Analysis

Portland, OR

September – December 2021

Presentations and Publications

Presentations:

Differentially Private Weighted Linear Regression. Reed College Mathematics Colloquium. April 2023.

Differentially Private Data Analysis via Edit Distance. Reed College Mathematics and Computer Science Colloquiums. October 2022.

Differentially Private Data Analysis via Edit Distance. Reed College Summer Research Poster Session. October 2022.

Coalescing Random Walks on an n -cycle. Reed College Mathematics Colloquium. September 2021.

Coalescing Random Walks on an n -cycle. 2021 REU Virtual Conference. August 2021.

Publications:

Zach Barbanell, Wolfgang Brightenburg, Kellen Brosnahan, Louise Xu, **Daniel Zou**, Andrew Bray, Adam Groce. *Differentially Private Data Analysis via Edit Distance*. (in preparation)

Skills

Computing: Python, R/RStudio (base and tidyverse), Basic C/C++

Publishing: $\LaTeX$, knitr

Other: Git/Github

Languages

English: Native Speaker

Chinese: Native Speaker